SINGAPORE DIMENSIONS • GRADE 4A • CHAPTER 5

Fractions: Mixed Numbers & Addition

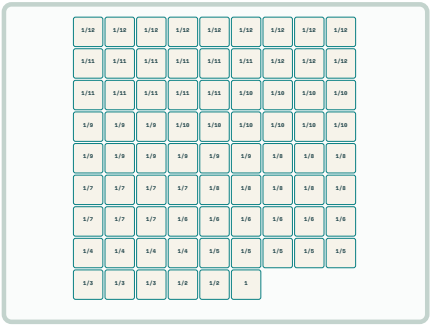
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.4.NF.B.3

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Fraction Blocks

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Stack two Whole blocks (84mm each) plus one 1/2 block. Add one 3/4 tower.

Sketch Model on Paper:
Draw shaded tape models for 2 1/2 and 3/4.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	2	Step 1	Execute [2]
2.	·	Step 2	Execute [.]
3.	1	Step 3	Execute [1]
4.	·	Step 4	Execute [.]

EXPECTED LCD X: 4 1/4 (or 17/4)

1 **Add 1 1/4 + 2 1/4.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Add 3 1/2 + 1 1/2.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Subtract 3 1/4 - 1 1/2.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Add 2 2/3 + 1 1/3.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 5

Fractions: Mixed Numbers & Addition • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Fraction Studio (84.0mm Whole + Unit Fraction)

CAD Blueprint: fraction_blocks

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/fraction_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Color-Coded 1/1 to 1/12 Miniblocks (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Add 1 1/4 + 2 1/4.** Standard: CCSS.MATH.4.NF.B.3	3 1/2	[1.1.4] [ENTER] [2.1.4] [+]
2	**Add 3 1/2 + 1 1/2.** Standard: CCSS.MATH.4.NF.B.3	5	[3.1.2] [ENTER] [1.1.2] [+]
3	**Subtract 3 1/4 - 1 1/2.** Standard: CCSS.MATH.4.NF.B.3	1 3/4	[3.1.4] [ENTER] [1.1.2] [-]
4	**Add 2 2/3 + 1 1/3.** Standard: CCSS.MATH.4.NF.B.3	4	[2.2.3] [ENTER] [1.1.3] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Emphasize that the first dot separates the whole number; the second separates numerator and denominator.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.